

Memory & Processors

Primary Memory vs. Secondary Storage

Computers use different areas to hold data depending on how fast the processor needs to access it.

- **Main Memory (Primary Storage):** High-speed memory that the processor can access directly and quickly. It holds data and instructions currently being processed (e.g., RAM and ROM).
- **Secondary Storage:** Slower, long-term storage where your files, games, and operating system live when not in use (e.g., Hard Disks and SSDs). Programs must be copied into main memory before they can run.

Random Access Memory (RAM)

RAM stands for **Random Access Memory**. When you launch an app or open a file, it is loaded into RAM from secondary storage so the processor can execute it.

- **Volatile:** RAM cannot store data without electrical power. Turning off the computer completely erases everything inside RAM.
- **Capacity Impact:** The more RAM you have, the more programs and large files you can run smoothly at the exact same time.

The Danger of Virtual Memory

When your computer runs completely out of physical RAM space, the Operating System creates **Virtual Memory** by using an area of your secondary storage.

- Because secondary storage is significantly slower than RAM, your system performance will drop noticeably.
- If your system uses a Solid State Drive (SSD), constantly swapping files back and forth between RAM and Virtual Memory can drastically reduce the lifespan and performance of that SSD.

Read Only Memory (ROM)

ROM stands for **Read Only Memory**. It is designed to store data permanently and securely.

- **Non-Volatile:** Data is safe and is **not lost** when the computer power is switched off.
- **Uses:**
 - *Single-Purpose Devices:* Stores the only program needed to run appliances like calculators, digital watches, or washing machines.

- *General-Purpose Computers*: Stores the **boot** (start-up) program that tells the computer how to turn on its hardware components and load the main Operating System.

The 4 Types of ROM

1. **Mask Programmed ROM**: Standard ROM that cannot be edited or changed after it leaves the factory floor.
2. **PROM (Programmable ROM)**: Manufactured blank, allowing a device creator to write data to it exactly **once**.
3. **EPROM (Erasable Programmable ROM)**: The contents can be completely erased, but only by exposing the chip to strong **Ultraviolet (UV) light**.
4. **EEPROM (Electrically Erasable Programmable ROM)**: Can be erased and rewritten electronically by applying a voltage to a specific pin. This allows **firmware** (software that controls hardware) to be updated without taking the chip out of the machine.

Flash Memory

Flash memory is a highly advanced type of **EEPROM**. Like ROM, it is entirely non-volatile.

- **Key Features**: It has absolutely no moving parts, boasts incredibly fast access times, and has very low power consumption.
- **Common Uses**: Because it saves battery power, it is ideal for SSDs inside portable laptops and smartphones. It is also the core technology inside USB thumb drives and camera SD cards.

Summary Checklist: RAM vs. ROM

Feature	Random Access Memory (RAM)	Read Only Memory (ROM)
Saves data when power is cut?	✗ No (Volatile)	Yes (Non-volatile)
Can swap instructions in/out?	Yes	✗ No
What does it store?	Temporary data in current use	Permanent start-up data/firmware
Can the user physically upgrade it?	Yes (Adding more sticks)	✗ No

Processors (CPUs)

The processor is built from one or more **Central Processing Units (CPUs)** that execute software instructions.

- **The Processor Cycle:** The loop of **fetching** an instruction from memory, **decoding** what actions are required, and **executing** those tasks.
- **Cores:** Each individual CPU inside a processor is called a **core**. A quad-core processor has four cores working at once, meaning it can handle up to four times as much work as a single-core chip.
- **The Energy Advantage:** Multi-core processors allow individual cores to run slower. This uses less power, generates less heat, keeps laptops quieter by turning down fans, and extends mobile battery life.

Measuring Processor Speed

Processor speed is measured by its clock speed—the number of **clock cycles per second** it can carry out. It is measured in Hertz (Hz):

- **Hertz (Hz):** 1 cycle per second.
- **Kilohertz (kHz):** 1,000 cycles per second.
- **Megahertz (MHz):** 1,000,000 cycles per second.
- **Gigahertz (GHz):** 1,000,000,000 cycles per second.

 **Exam Trap:** A higher Gigahertz rating doesn't *always* mean a computer finishes a task first. Different processor designs allow some CPUs to complete significantly more work/instructions during a single clock cycle than others.

CHAPTER QUESTIONS

1. State what is meant by 'ROM'. (1)
2. State what is meant by 'RAM'. (1)
3. Explain why increasing the amount of RAM in a system improves the user experience. (3)
4. Describe what virtual memory is and why it is used. (2)
5. List three characteristics of RAM that make it different to ROM. (3)
6. Explain two reasons why the characteristics of flash memory make it suitable for use as secondary storage in smartphones. (4)
7. Which one of these describes the term 'clock cycle'? (1)
 - a. The number of times per second that a processor can carry out instructions.
 - b. The number of instructions that a processor can carry out each second.
 - c. The number of times that a processor can load instructions for processing.
 - d. The number of times that each process repeats in a processor.